

T-Coil Design with Embedded Inductor

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Project Overview

- Goal: Design t-coil with PCB-embedded inductor to compensate for heavy load capacitance and enhance channel bandwidth
- Outline:
 - T-coil background
 - Embedded inductor design and simulation
 - PCB design
 - Bridge capacitor value selection
 - Lab characterization and comparison with simulations

T-Coil Overview

Bridged T-Coil [1]

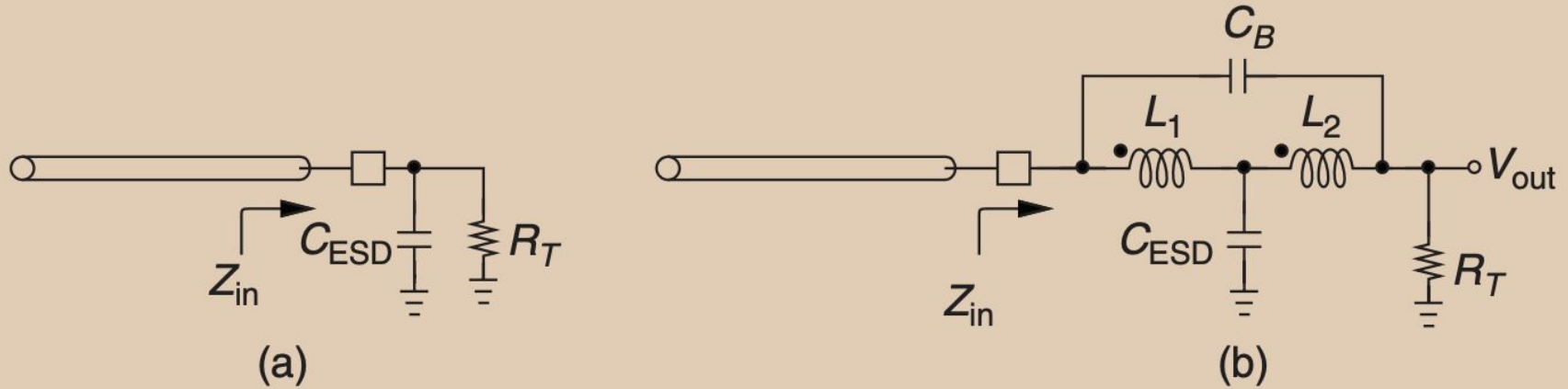
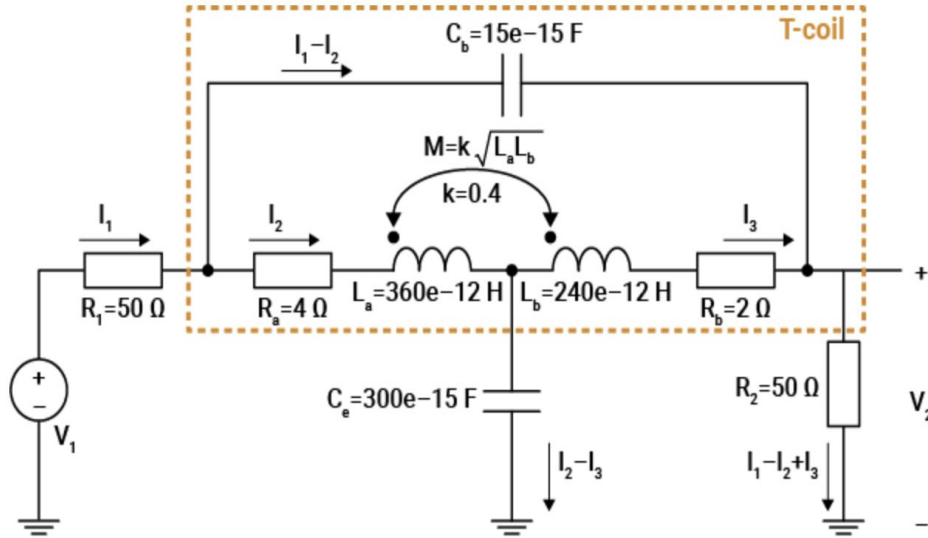


FIGURE 4: (a) An input network with an ESD device and (b) an input network using a T-coil for broadband matching.

Formulas for determining L, k, and C_B [1]



$$L_1 = L_2 = \frac{R_D^2 C_L}{4} \left(1 + \frac{1}{4\zeta^2} \right) \quad (13)$$

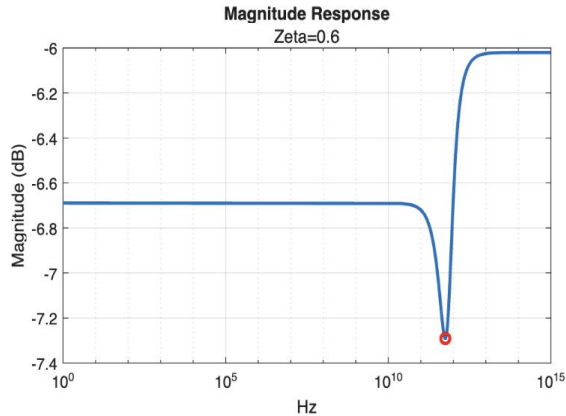
$$k = \frac{4\zeta^2 - 1}{4\zeta^2 + 1} \quad (14)$$

$$C_B = \frac{C_L}{16\zeta^2}, \quad (15)$$

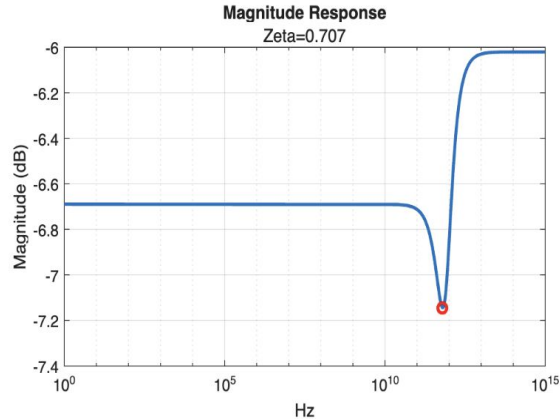
MATLAB Simulation

Selected $C_L = 4 \text{ pF}$ to mimic ESD diode capacitance (e.g. Ethernet, USB 2.0) [2]

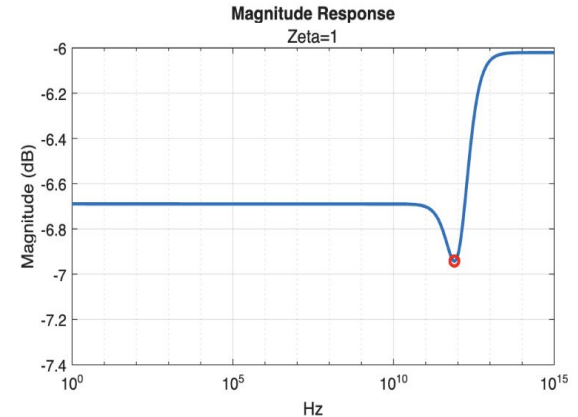
Dip becomes smaller as ζ increases. Resonant frequency also increases with ζ .



$f_{\text{res}} = 575.01 \text{ GHz}$



$f_{\text{res}} = 625.02 \text{ GHz}$



$f_{\text{res}} = 775.02 \text{ GHz}$

Calculation results

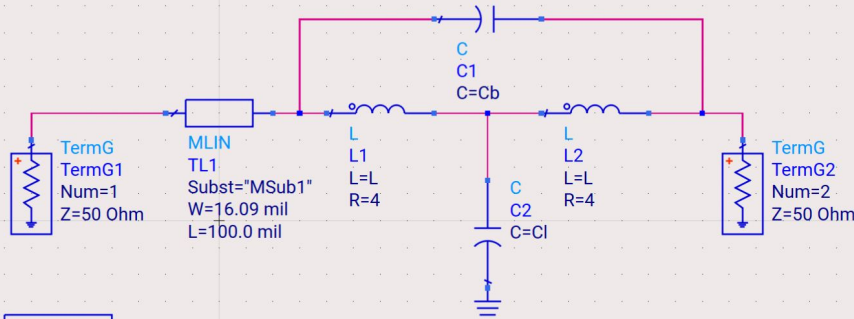
Updated with $R_D = 50$ Ohms, $C_L = 4$ pF

Zeta	L	k	C_B	f_res (Hz)
0.6	4.2361e-09 H	0.18033	6.9444e-13 F	5.7501e+11
0.707 ($\sqrt{2}/2$)	3.7504e-09 H	0.3332	5.0015e-13 F	6.2502e+11
1	3.125e-09 H	0.6	2.5e-13 F	7.7502e+11

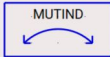
As Zeta (damping factor) increases:

- k (coupling between inductors) increases
- L and C_B decreases

Keysight ADS Simulation



Thickness	Layer
0.6 mil (0.0152mm)	silkscreen
0.6 mil (0.0152mm)	solder resist
1.7 mil (0.0432mm)	1 oz copper clad+plated
7.87 mil (0.1999mm)	FR408HR 2113 prepreg



Mutual
Mutual1
K=k
M=
Inductor1="L1"
Inductor2="L2"



S-PARAMETERS

S_Param
SP1
Start=1 Hz
Stop=10 GHz
Step=



VAR
VAR1
 $k = \frac{4 \cdot (D^{**2}) - 1}{4 \cdot (D^{**2}) + 1}$
 $C_b = C_l / (16 \cdot (D^{**2}))$
 $L = ((R_d^{**2}) \cdot C_l / 4) \cdot (1 + (1 / (4 \cdot (D^{**2}))))$
Rd=50
Cl=4e-12
D=1 {-t} {o}

MSub

MSUB
MSub1
H=7.87 mil
Er=3.61
Mur=1
Cond=5.96E+7
Hu=1e+33 mm
T=1.7 mil
TanD=0.0098
Rough=0 mm
Bbase=
Dpeaks=

GOAL

Goal
OptimGoal1
Expr="dB(S(2,1))"
SimInstanceName="SP1"
Weight=1



OPTIM

Optim
Optim1
OptimType=Random
UseAllOptVars=yes
UseAllGoals=yes
SaveAllTrials=no

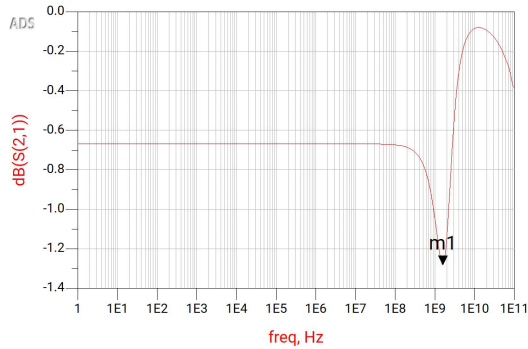
GOAL

Goal
OptimGoal2
Expr="dB(S(1,1))"
SimInstanceName="SP1"
Weight=1

Keysight ADS Simulation

Zeta = 0.6

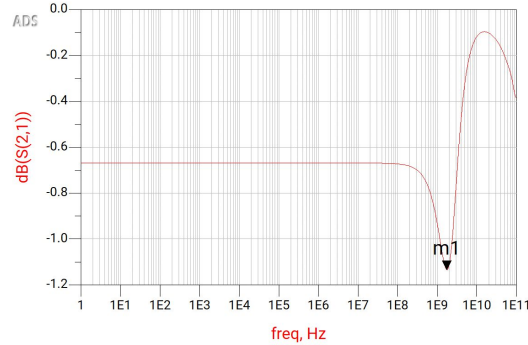
m1
freq=1.585 GHz
dB(S(2,1))=-1.281



f_{res} = 1.585 GHz

Zeta = 0.707

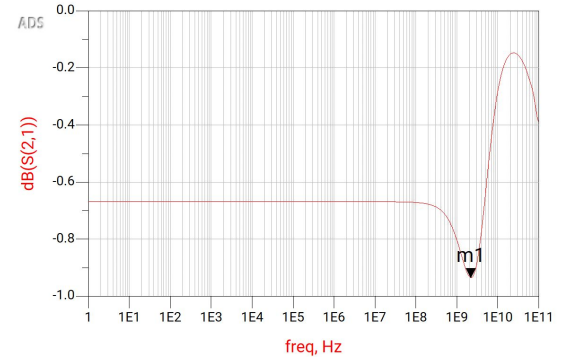
m1
freq=1.738 GHz
dB(S(2,1))=-1.134



f_{res} = 1.738 GHz

Zeta = 1

m1
freq=2.188 GHz
dB(S(2,1))=-0.935

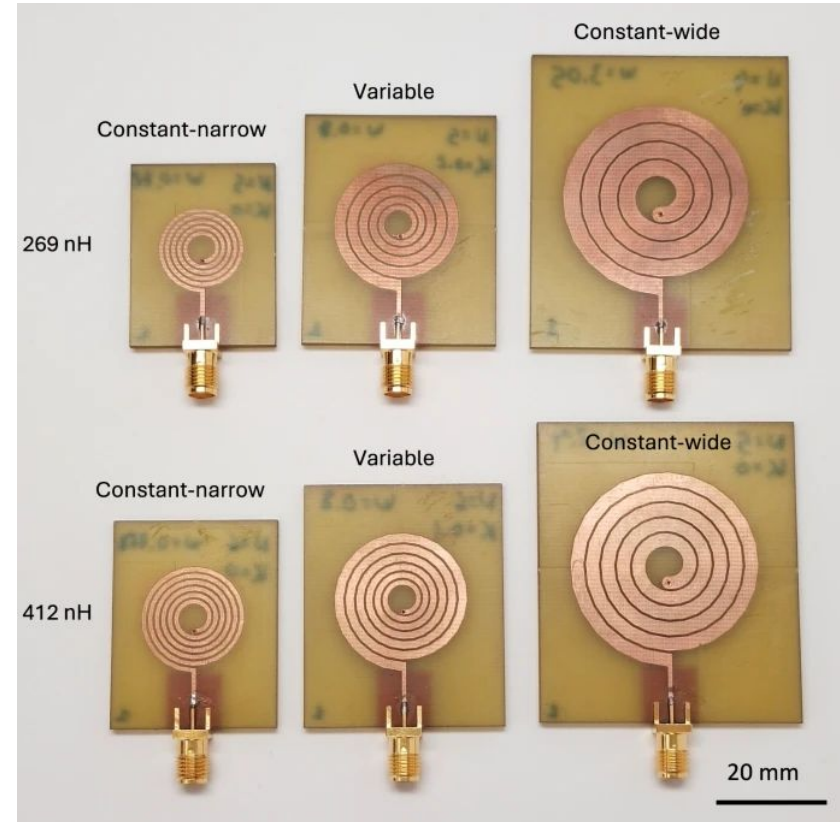


f_{res} = 2.188 GHz

Spiral Inductor

Spiral Inductor Paper [3]

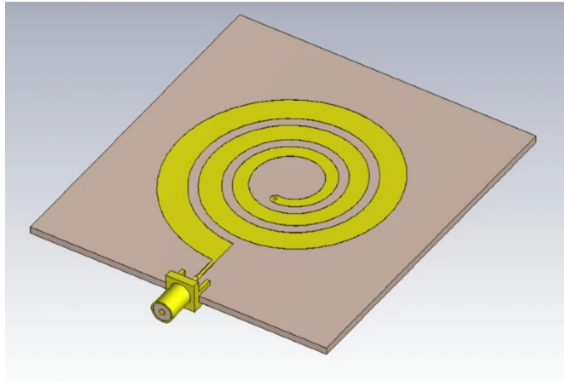
- Spiral planar inductors are used for Microwave applications.
- Planar spiral inductors are limited by two parasitics: series resistance R_s (loss) and inter-turn capacitance C_p (bandwidth / self-resonance).
- S-Parameters were measured between 1MHz and 500 MHz.



Geometry

- Circular geometry supports better current-path symmetry
- The following equation determines the polar width given the w_0 (initial width) and growth factor g (how much width increases per turn)

$$w(\theta) = w_0 \left(1 + g \frac{\theta}{2\pi} \right)$$

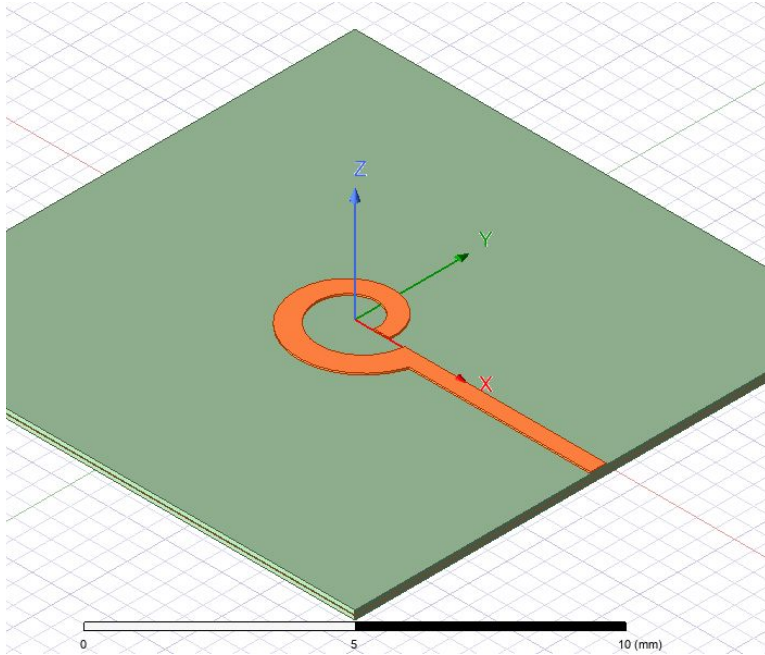


Fabrication

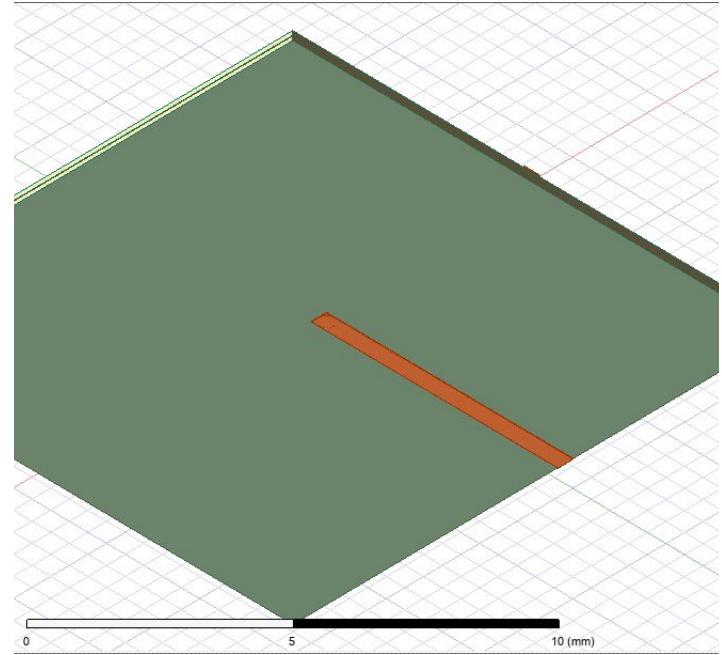
- Prototypes made on double-sided FR4 with 18 μm copper, patterned using a laser prototyping system
- Measured S_{11} from 1–500 MHz and from it computed equivalent inductance, equivalent impedance, and quality factor.

$$L_{eq}(f) = \frac{\text{Imag}(Z_{eq})}{2\pi f} \quad Q(f) = \frac{\text{Imag}(Z_{eq})}{\text{Real}(Z_{eq})} \quad Z_{eq}(f) = \frac{1 + S_{11}(f)}{1 - S_{11}(f)}$$

Spiral Inductor HFSS Design

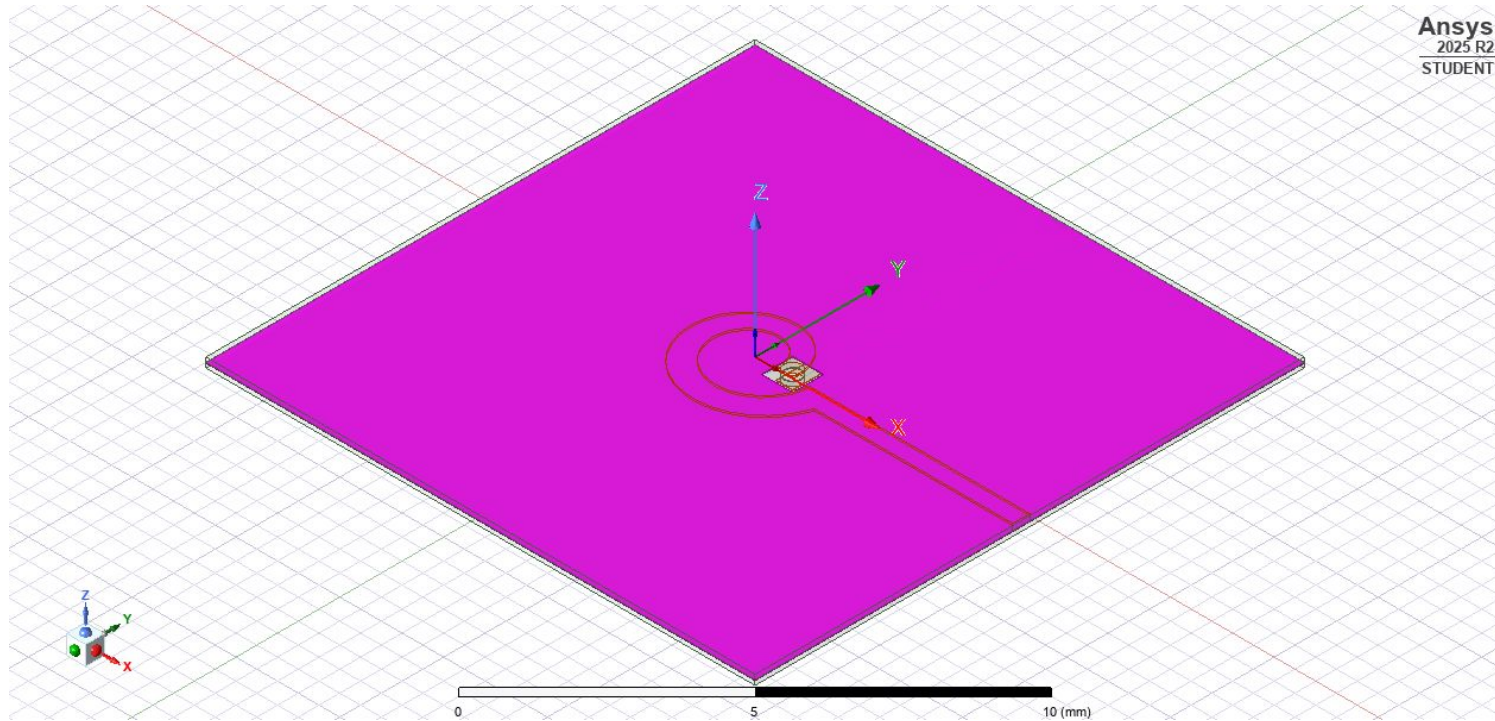


Front side



Back side

Spiral Inductor HFSS Design



Ground plane

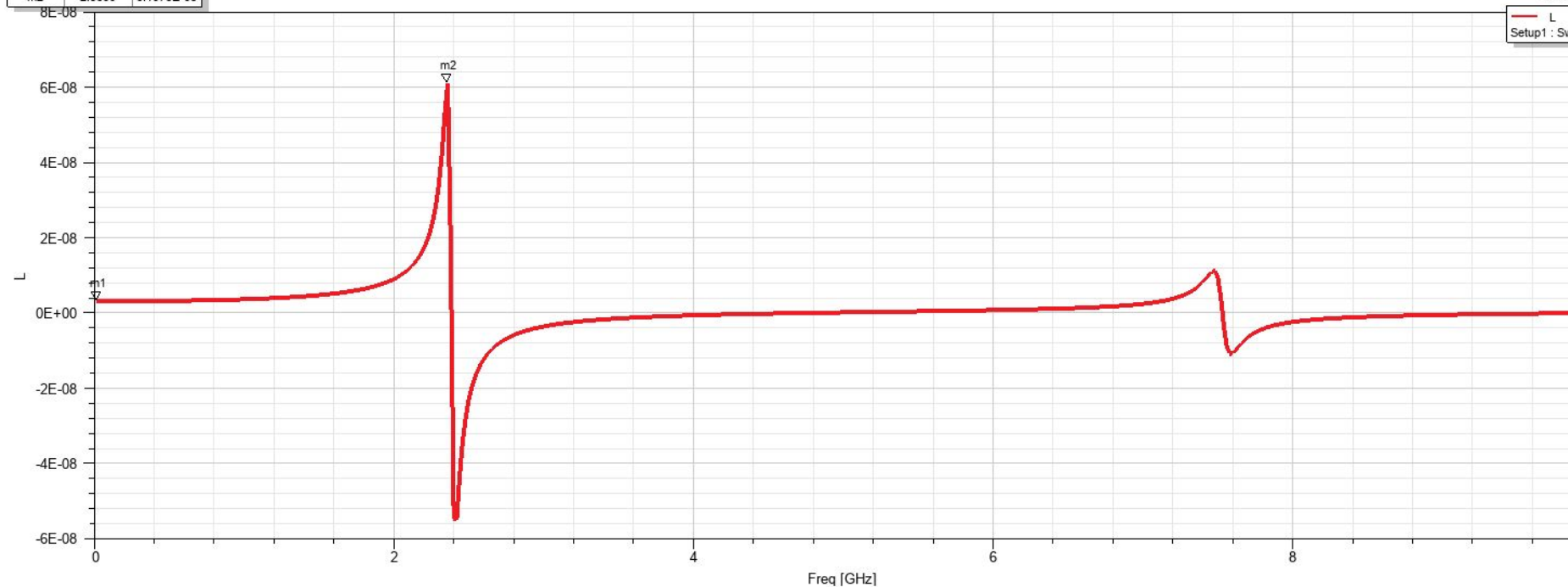
Spiral Inductor HFSS Simulation

Name	X [GHz]	Y
m1	0.0200	3.3727E-09
m2	2.3600	6.1076E-08

Inductance

HFSSDesign4

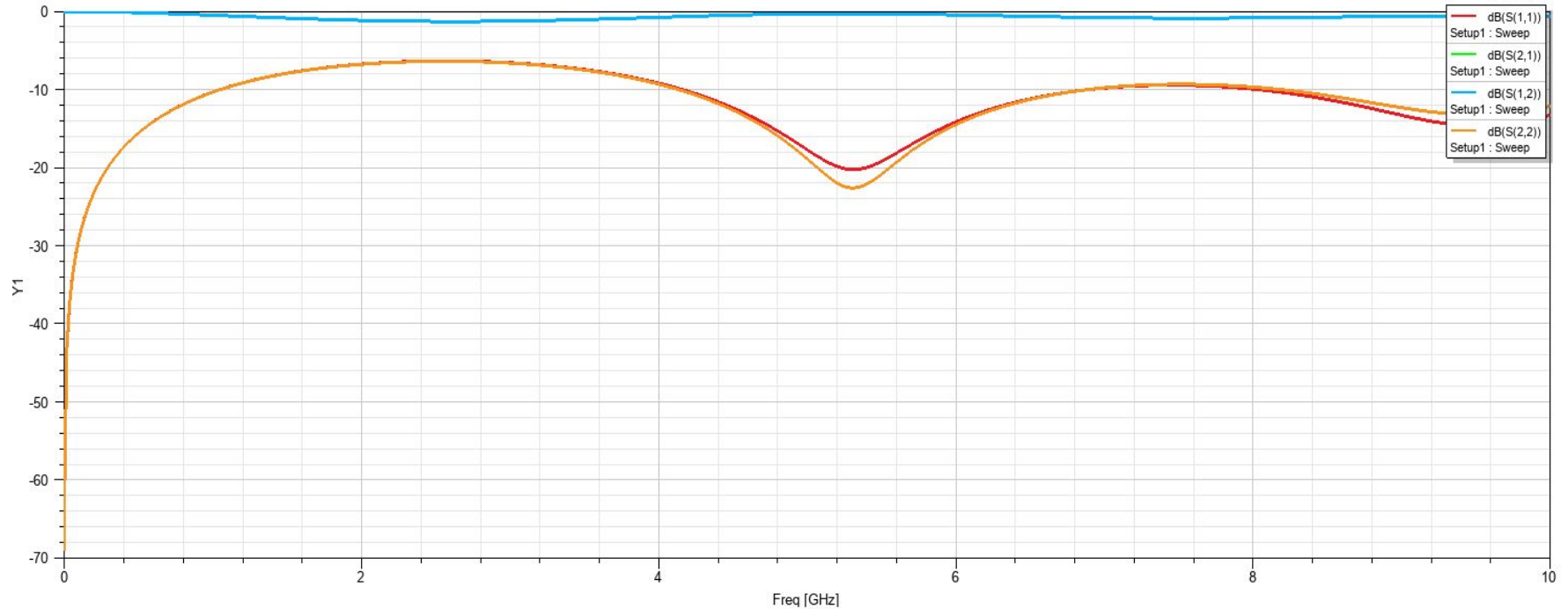
An
20
STU



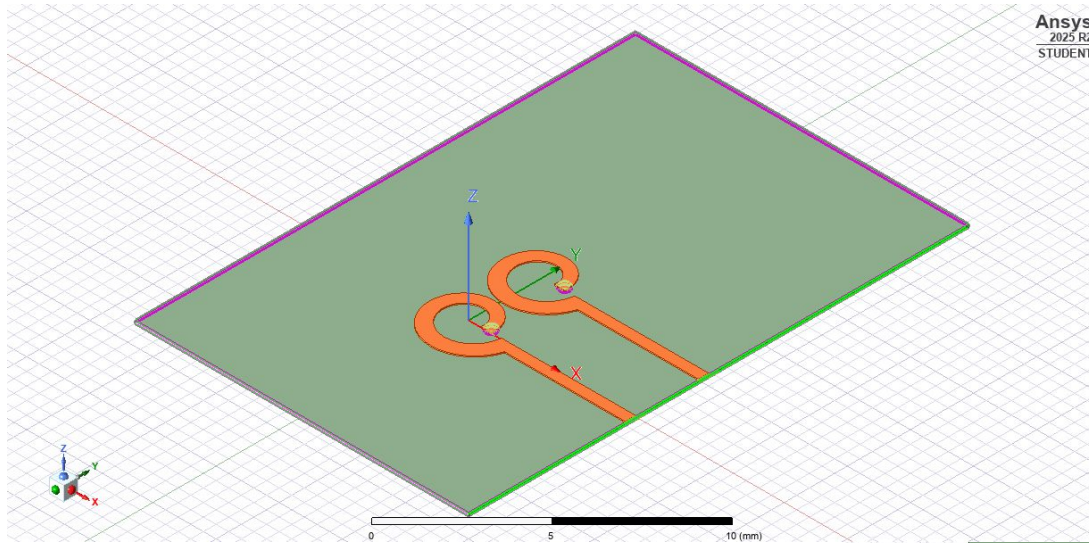
Spiral Inductor HFSS Simulation

S Parameter Plot1

HFSSDesign4
Ansys
2025 R2
STUDENT

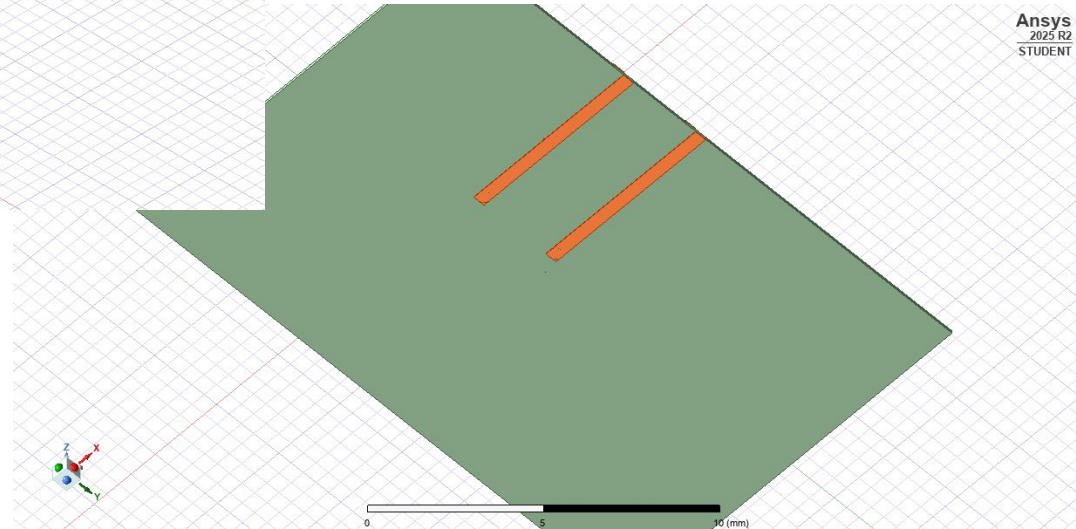


Spiral Inductor Mutual Inductance

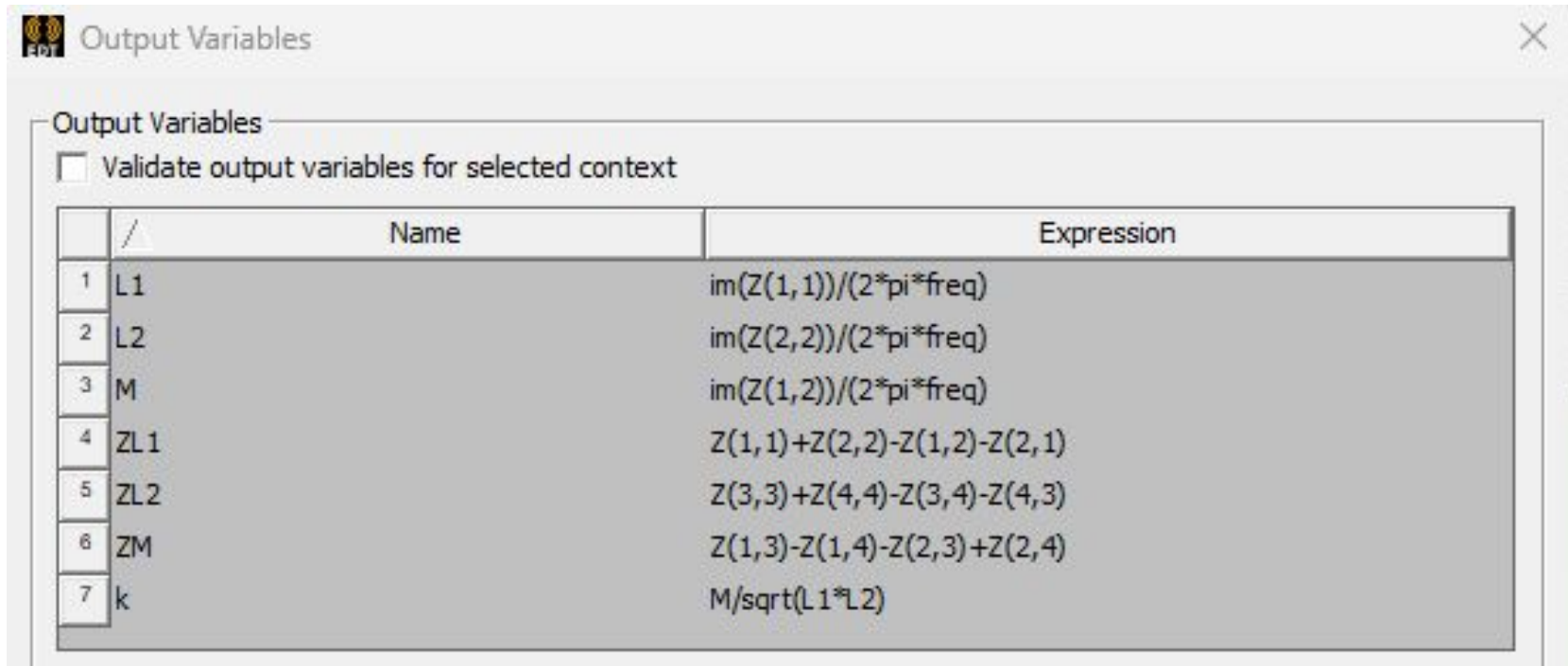


Front side

Back side



Spiral Inductor Mutual Inductance



	Name	Expression
1	L1	$\text{im}(Z(1,1))/(2*\pi*\text{freq})$
2	L2	$\text{im}(Z(2,2))/(2*\pi*\text{freq})$
3	M	$\text{im}(Z(1,2))/(2*\pi*\text{freq})$
4	ZL1	$Z(1,1)+Z(2,2)-Z(1,2)-Z(2,1)$
5	ZL2	$Z(3,3)+Z(4,4)-Z(3,4)-Z(4,3)$
6	ZM	$Z(1,3)-Z(1,4)-Z(2,3)+Z(2,4)$
7	k	$M/\text{sqrt}(L1*L2)$

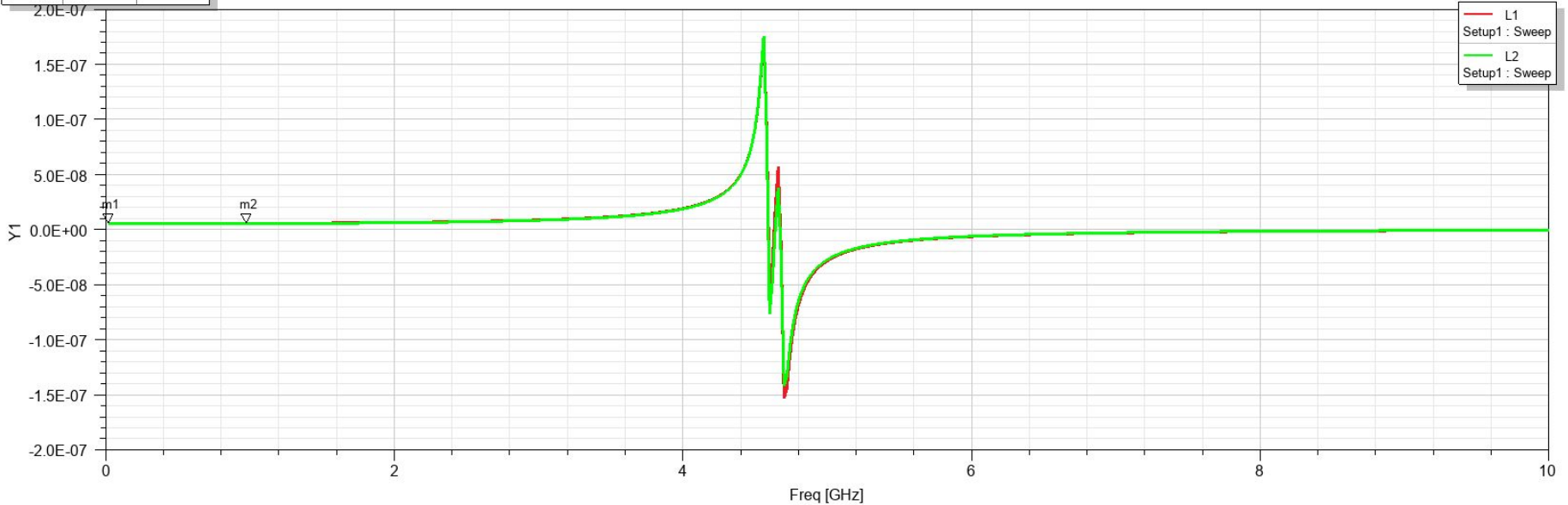
Formulas used for calculating inductance and coupling

Spiral Inductor Mutual Inductance

Name	X [GHz]	Y
m1	0.0200	5.4761E-09
m2	0.9800	5.5288E-09

Output Variables Plot1

no_ground
Ansys
2025 R2
STUDENT



Spiral Inductor Mutual Inductance

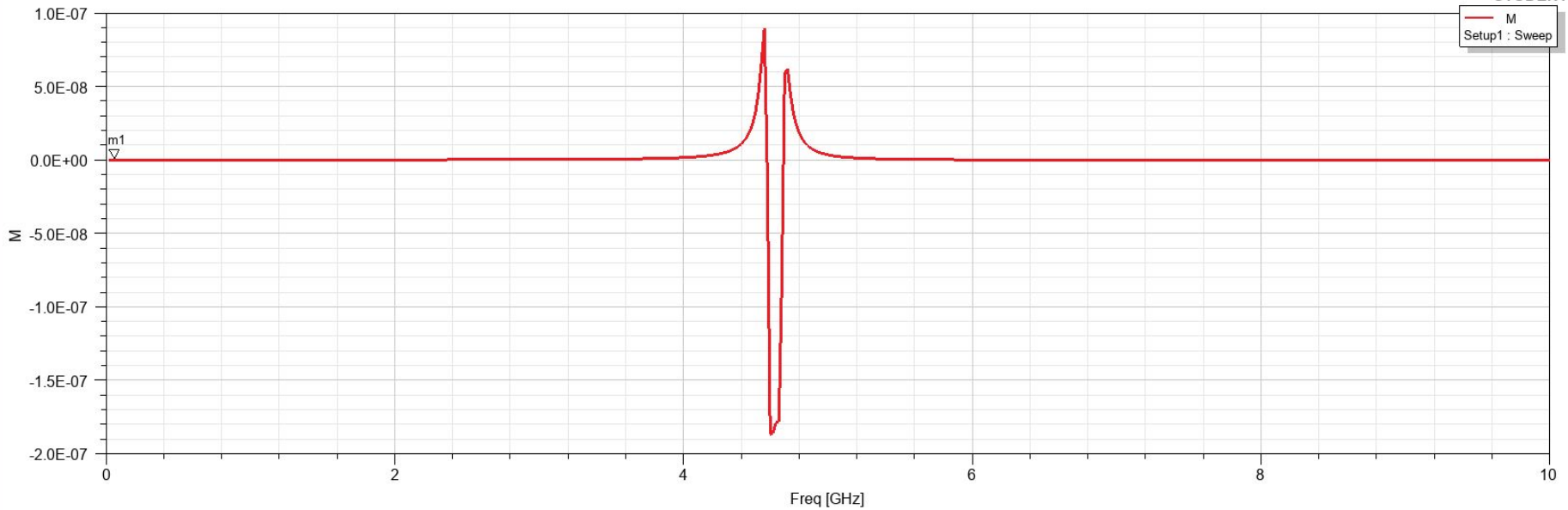
Name	X [GHz]	Y
m1	0.0600	1.2454E-10

Output Variables Plot2

no_ground

Ansys
2025 R2
STUDENT

M
Setup1 : Sweep

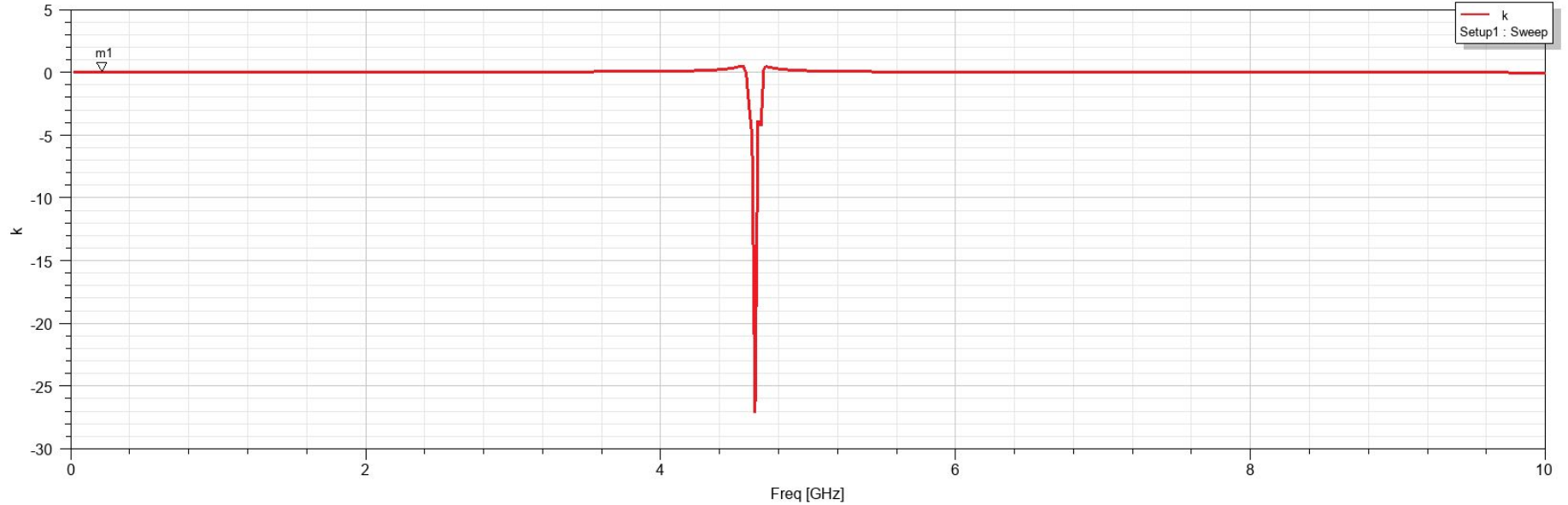


Spiral Inductor Mutual Inductance

Name	X [GHz]	Y
m1	0.2200	0.0230

Output Variables Plot3

no_ground
Ansys
2025 R2
STUDENT



Spiral Inductor

Old spiral design is not optimal for production:

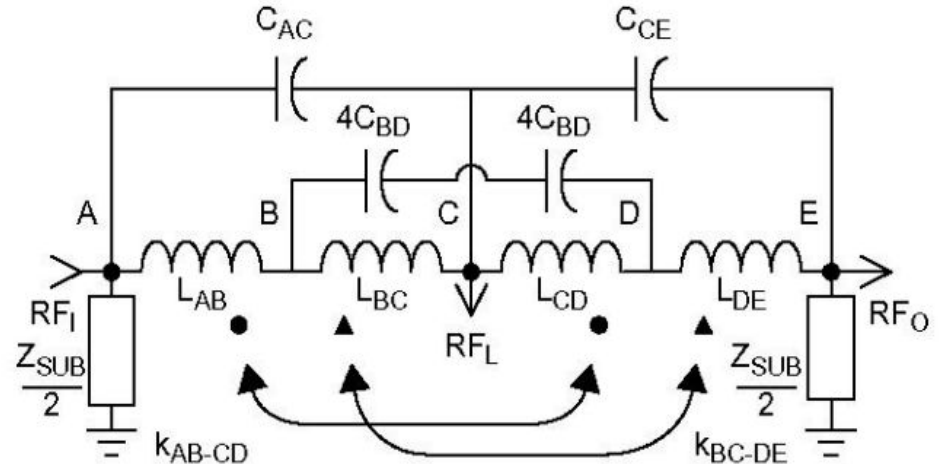
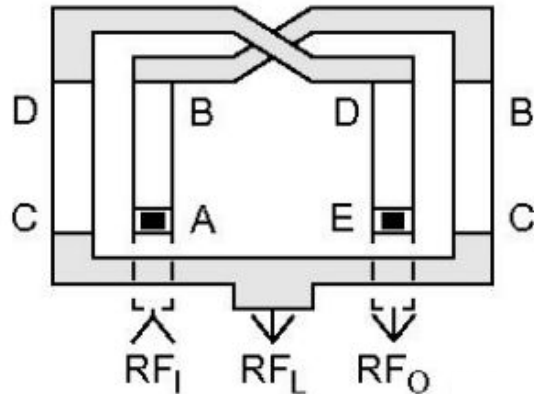
- Mutual inductance is too small (around $k=0.02$ instead of the target $k=0.6$)
- Inductors can be stacked on parallel planes to increase mutual inductance, but the load capacitor's placement will be tricky

Symmetric Inductor

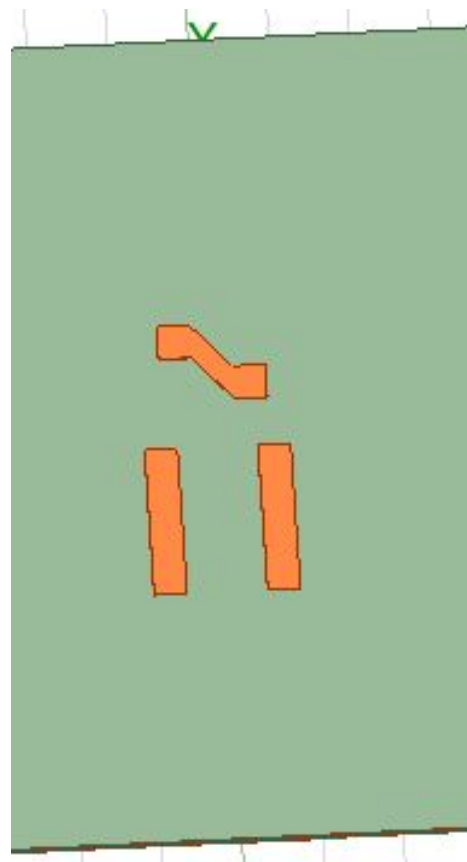
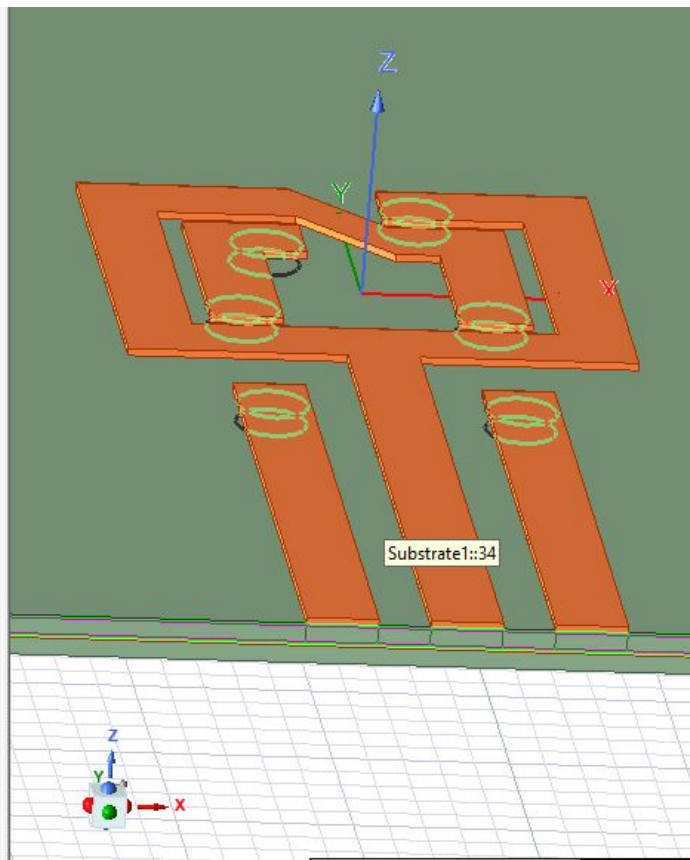
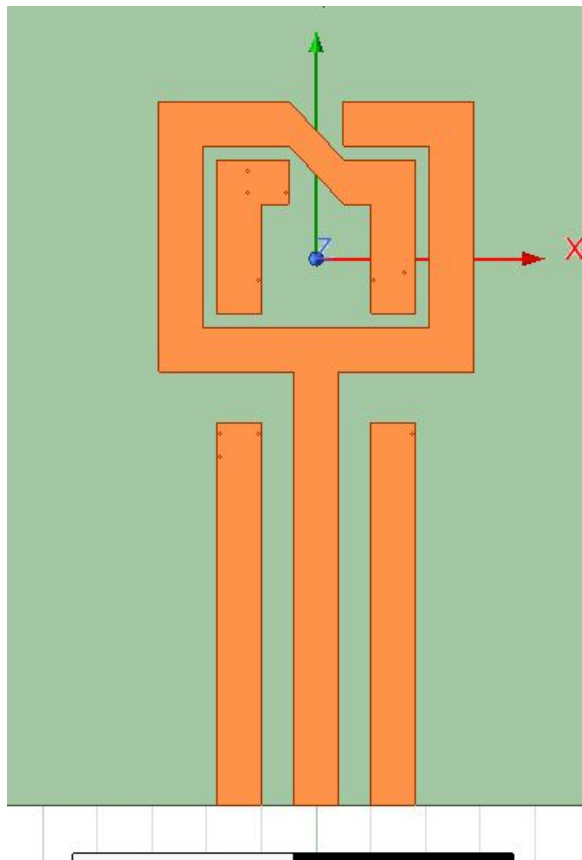
Symmetric T-Coil Paper [4]

New design inspired by paper about symmetric T-coils embedded in high-resistivity silicon

- Two inductors wrapped around each other, allowing for more coupling
- Planar design easier for layout



Symmetric T-Coil Design

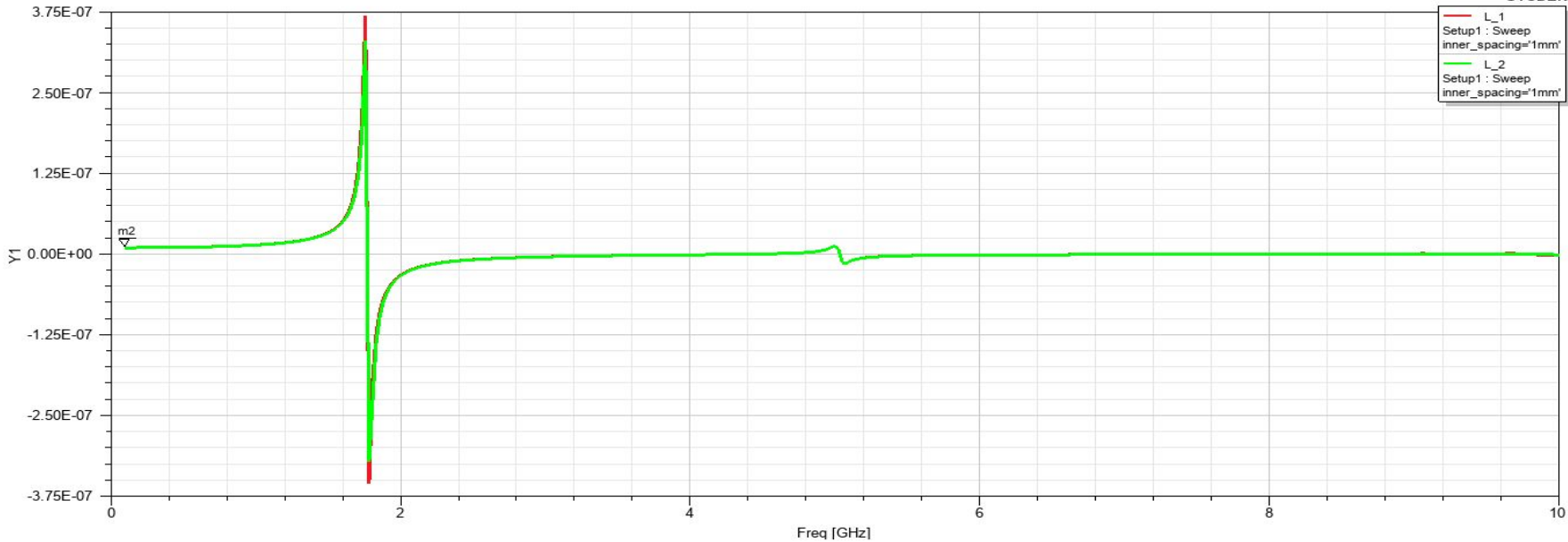


Symmetric T-Coil Design

Name	X [GHz]	Y
m2	0.1000	9.9984E-09

Inductance

HFSSDesign2
Ansys
2025 R2
STUDENT

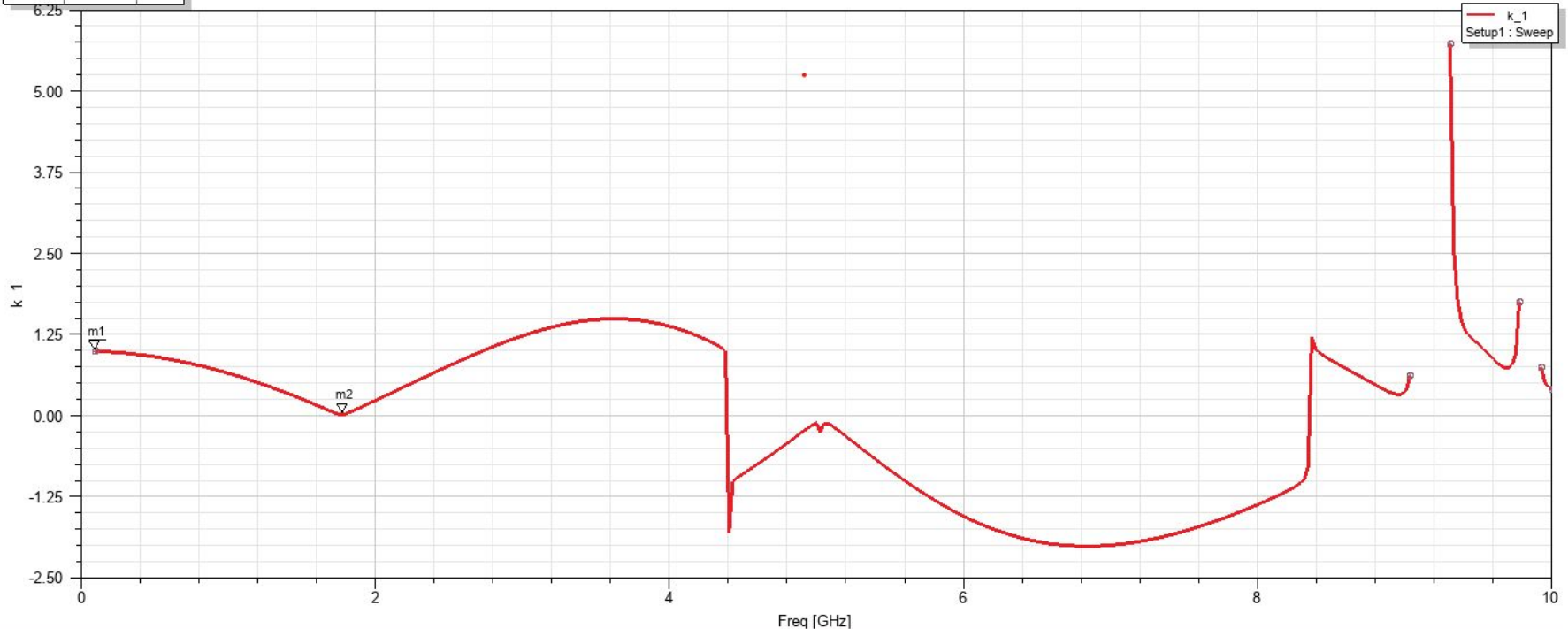


Symmetric T-Coil Design

Name	X [GHz]	Y
m1	0.1000	0.9964
m2	1.7830	0.0244

Coupling Coefficient

HFSSDesign2
Ansys
2025 R2
STUDENT

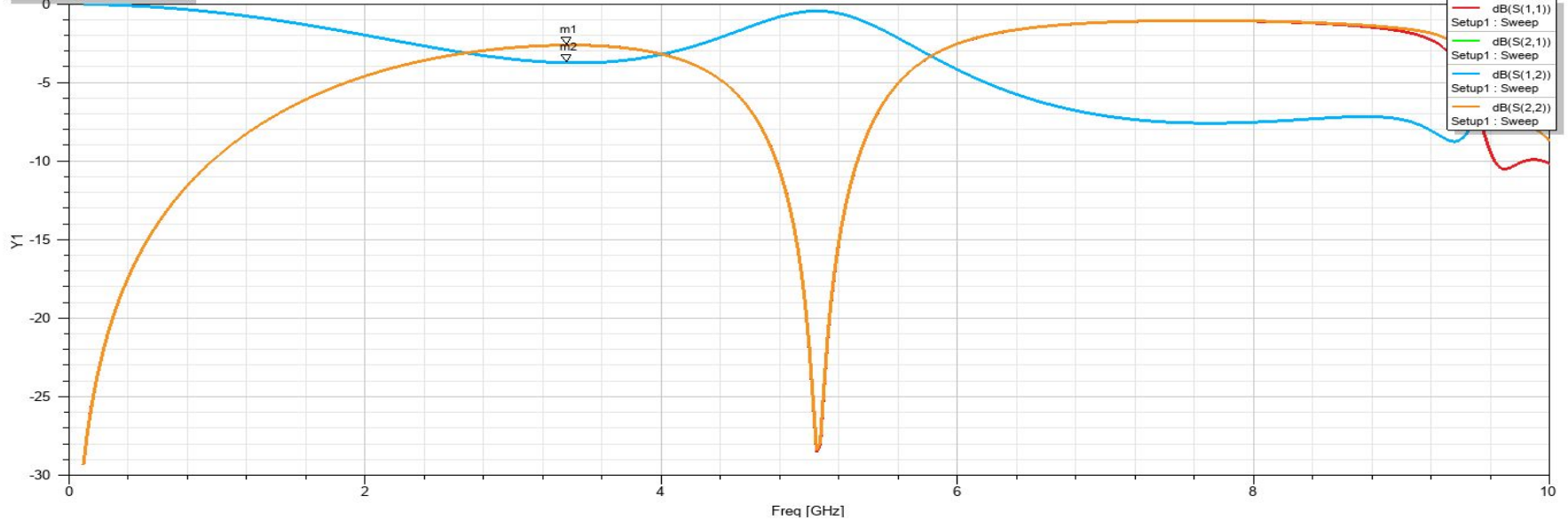


Symmetric T-Coil Design

Name	X [GHz]	Y
m1	3.3670	-2.6287
m2	3.3670	-3.7514

S Parameter Plot1

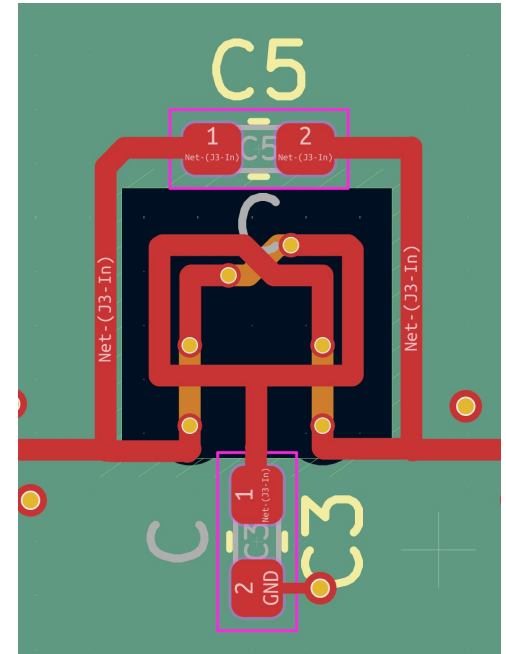
HFSSDesign2
Ansys
2025 R2
STUDENT

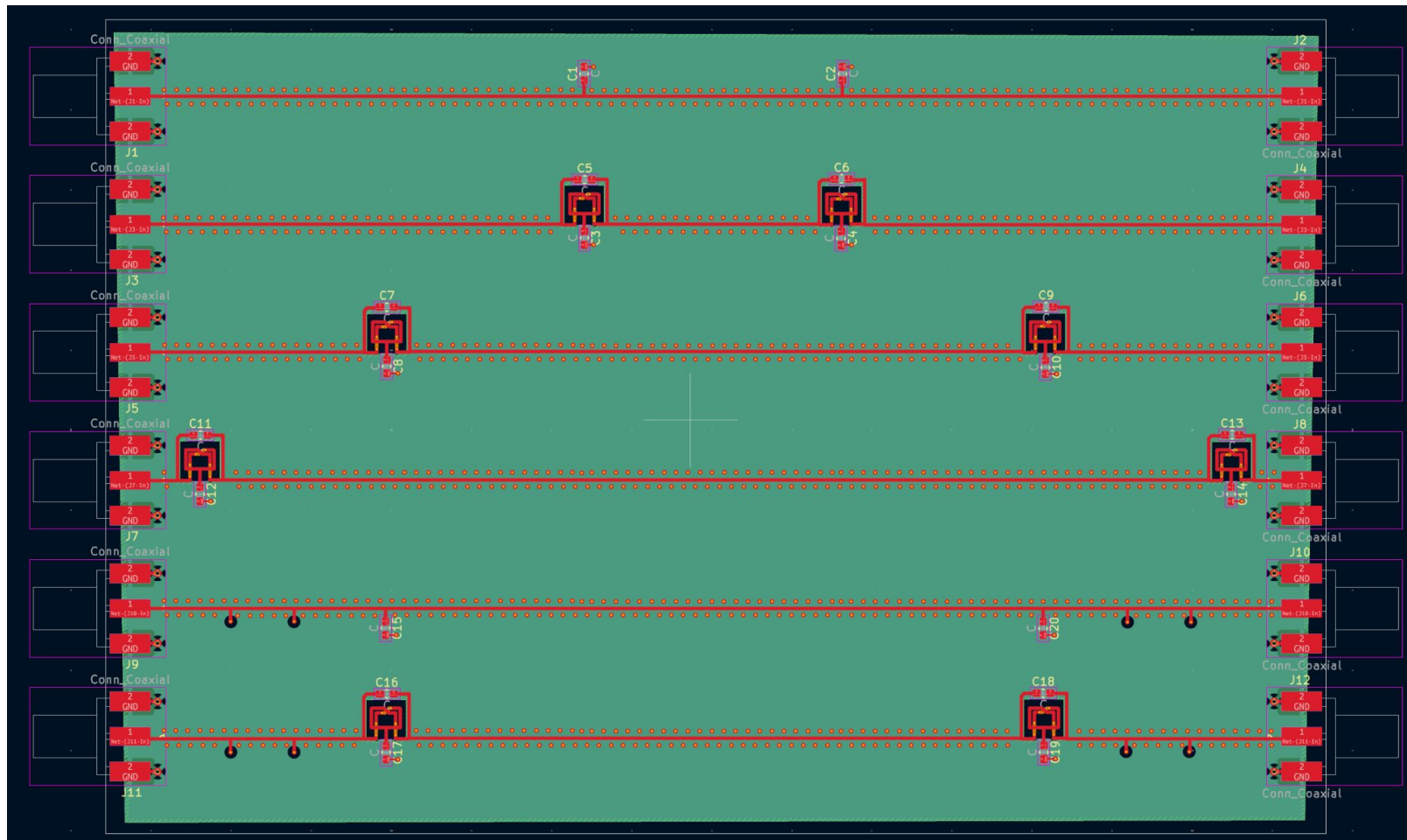


PCB Design and Simulation

PCB Design

- Inspired by design presented by Dr Walker Turner in which T-coils were used to compensate for parasitic capacitances on either side of package channel
- 6 channels:
 - Parasitic capacitances without T-coils
 - 1-inch channel with T-coils
 - 3-inch channel with T-coils
 - 5-inch channel with T-coils
 - Parasitic capacitances and vias without T-coils
 - Parasitic capacitances and vias with T-coils
- Via fencing used to shield channels





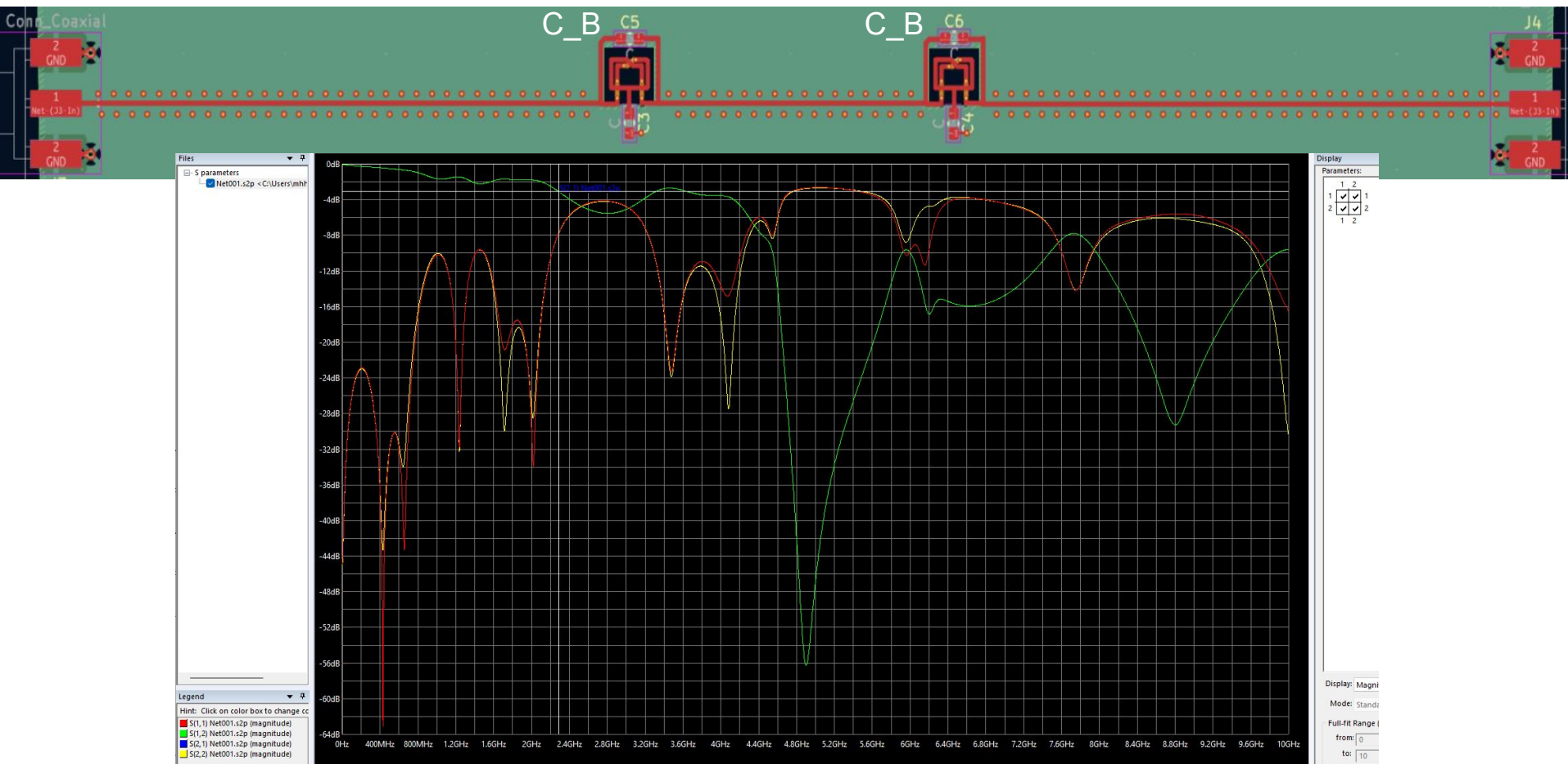
PCB Layout in KiCAD

Selecting Bridge Capacitor Values

- Exported ODB++ to HyperLynx and ran 3D extraction on T-coils and vias
- Simulated each channel with different bridge capacitor values
- 0.8pF consistently resulted in best channel bandwidth, measured by -3dB point
- Data below is for the one-inch channel; the same trend is seen in other channels

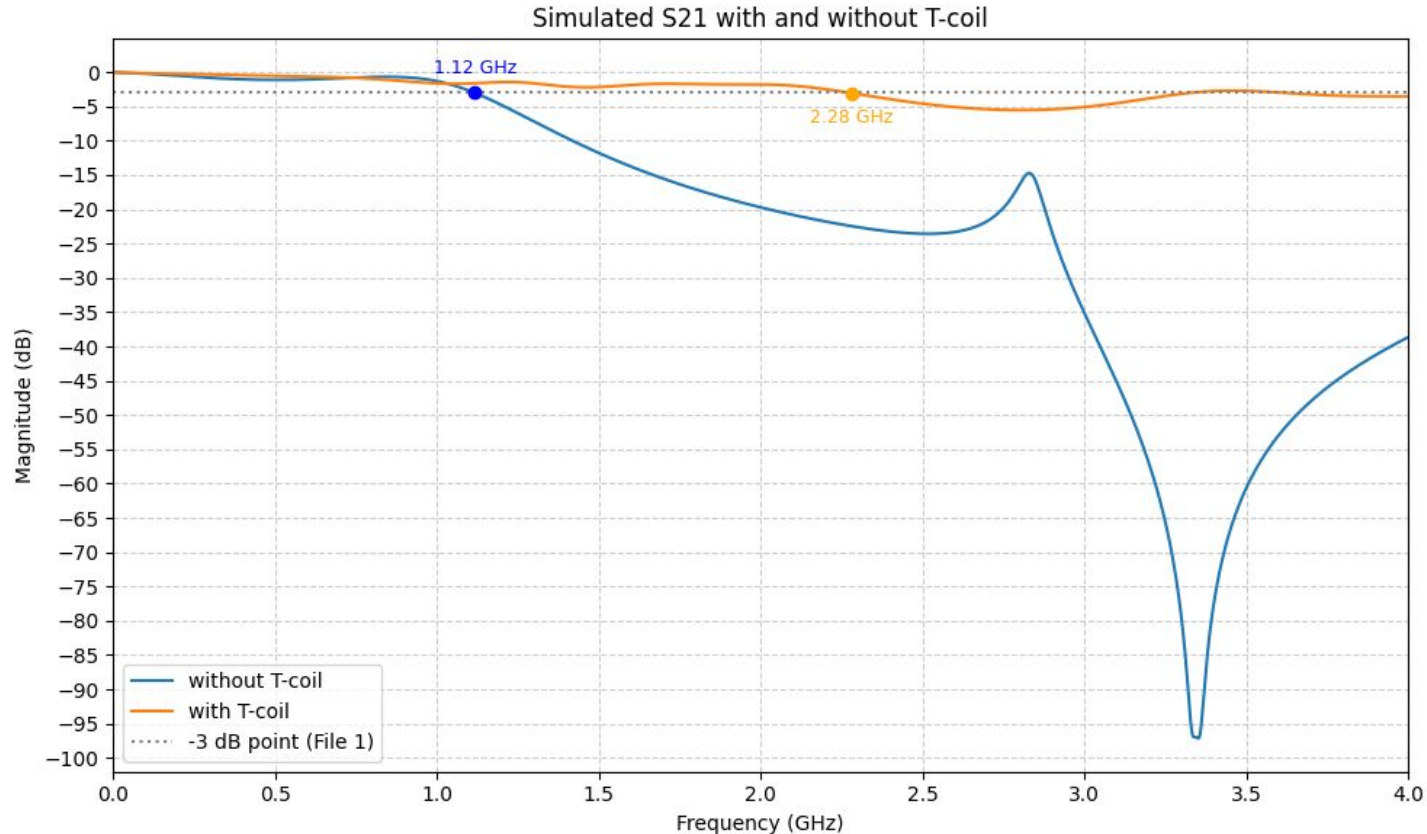
C_B (pF)	-3dB frequency (GHz)
0.5	1.657
0.6	2.214
0.7	2.243
0.8	2.286
0.9	1.421
1	1.307

Selecting Bridge Capacitor Values

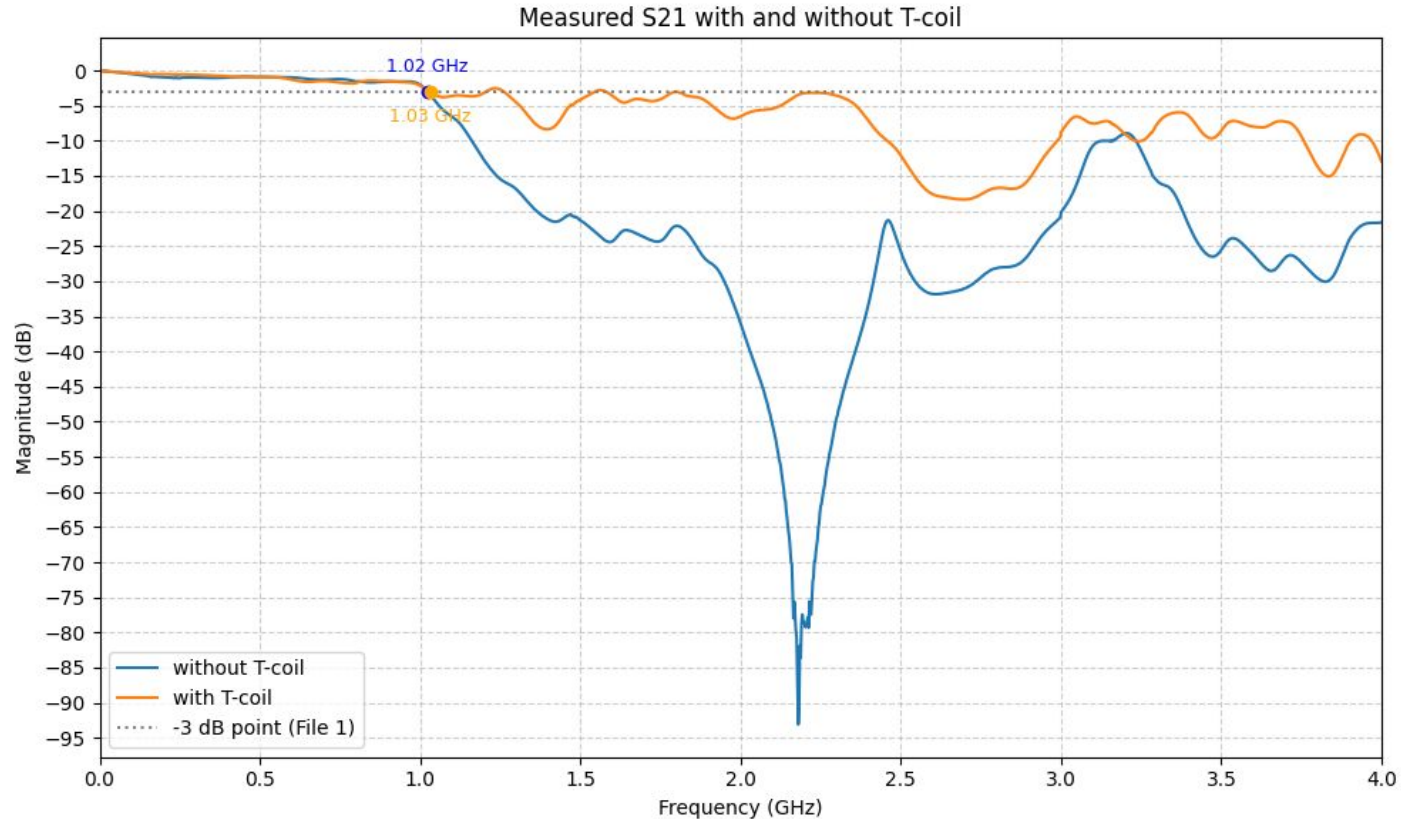


Lab Characterization

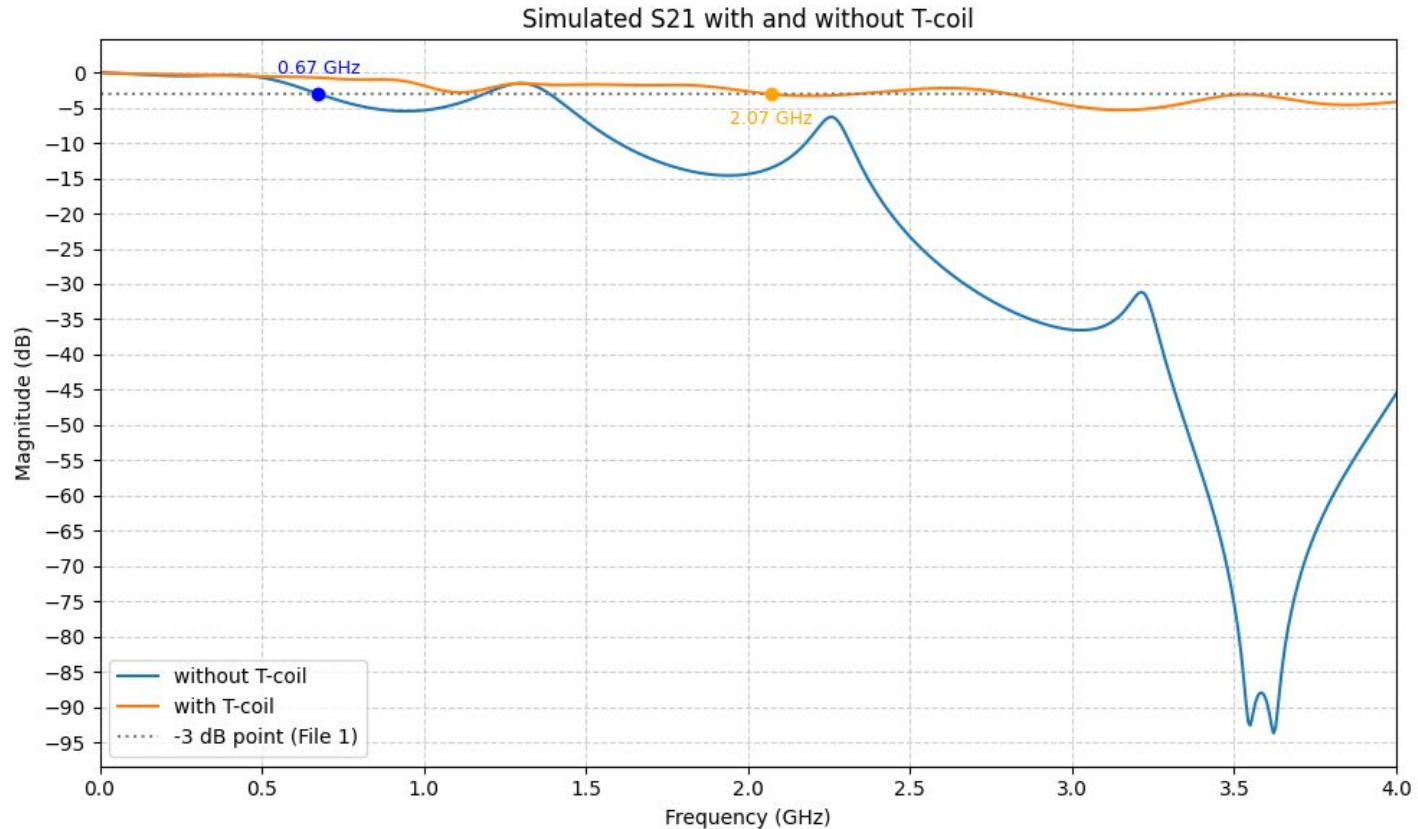
Parasitic Capacitance with and without T-coils (Simulated)



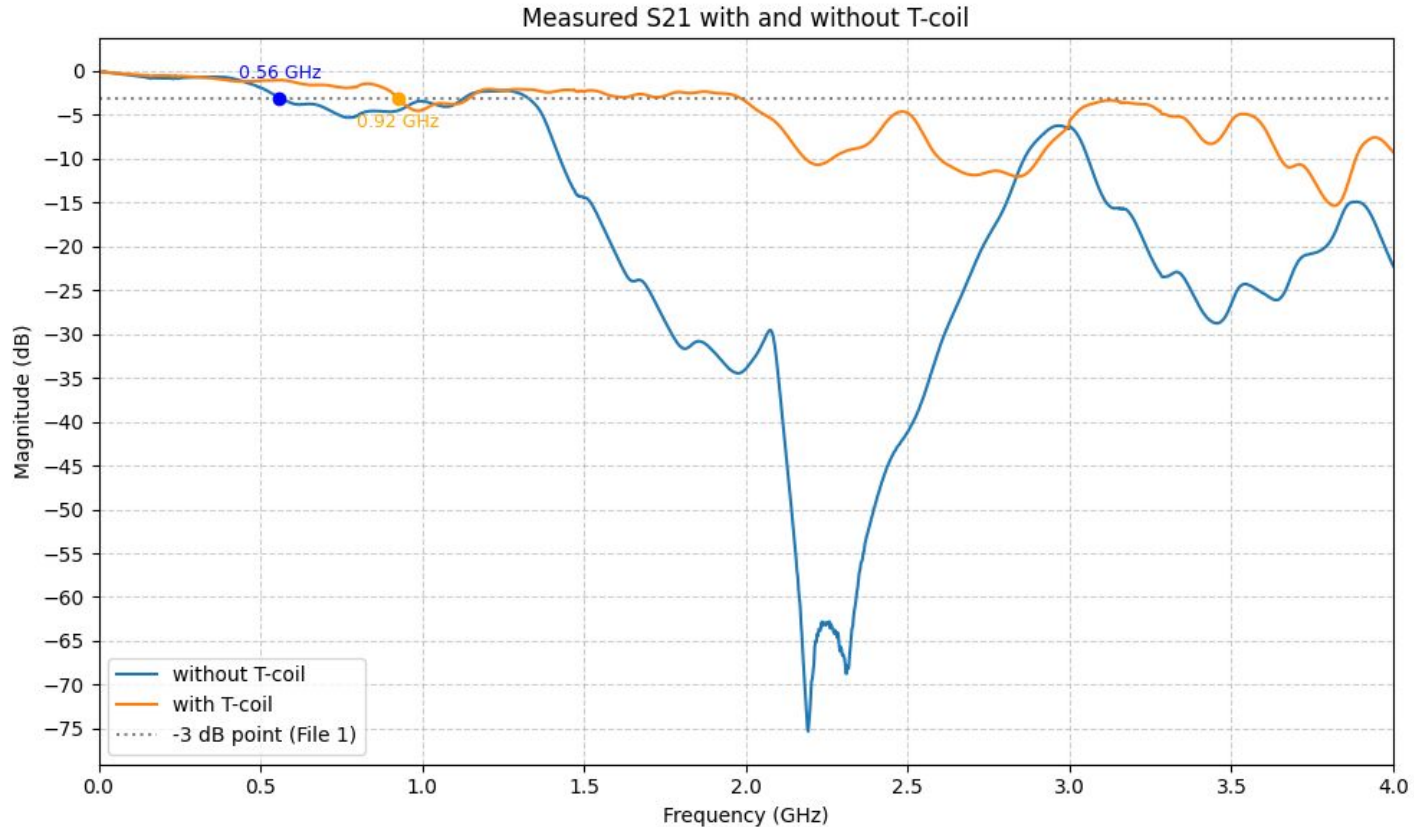
Parasitic Capacitance with and without T-coils (Measured)



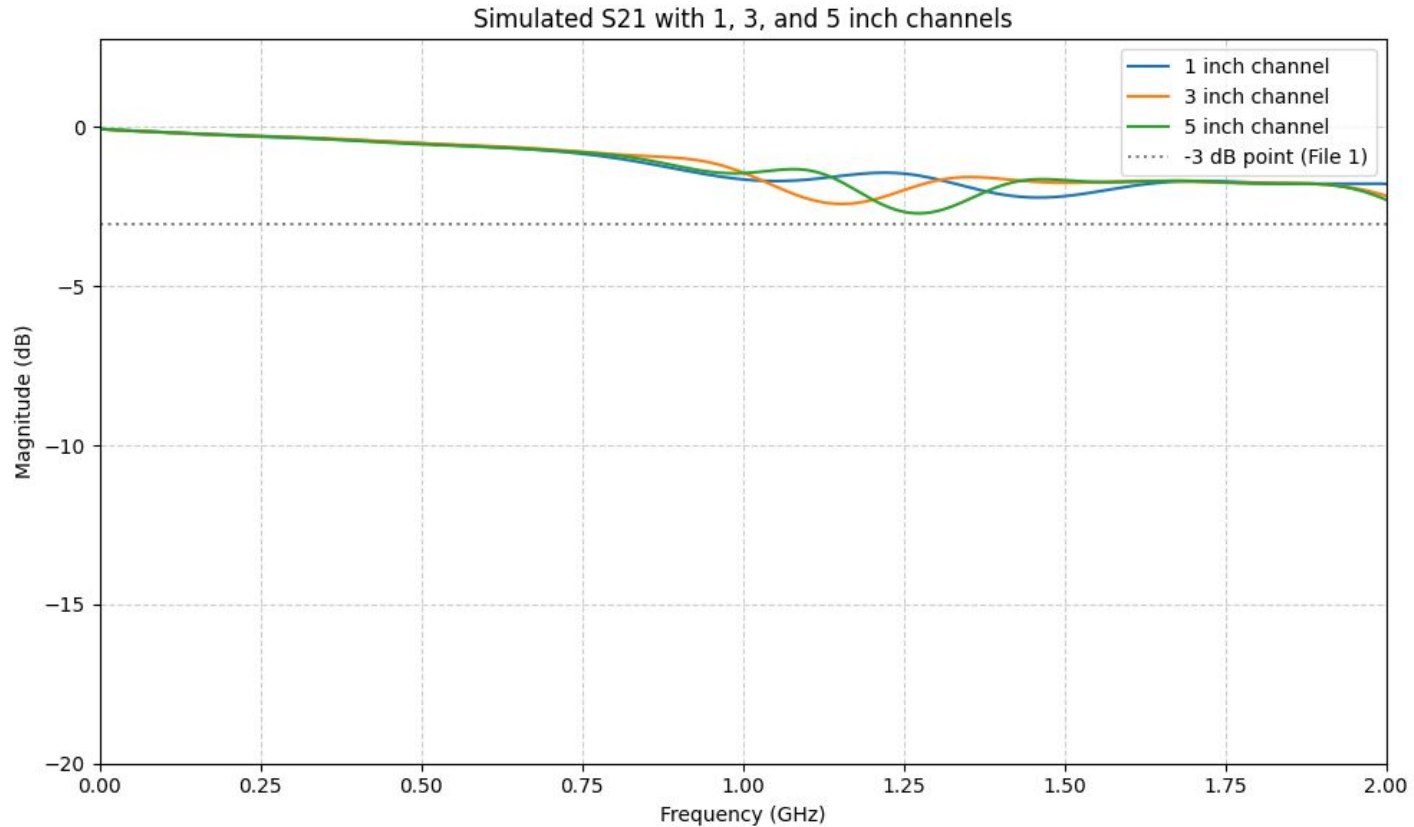
Parasitic Cap and Vias with and without T-coils (Simulated)



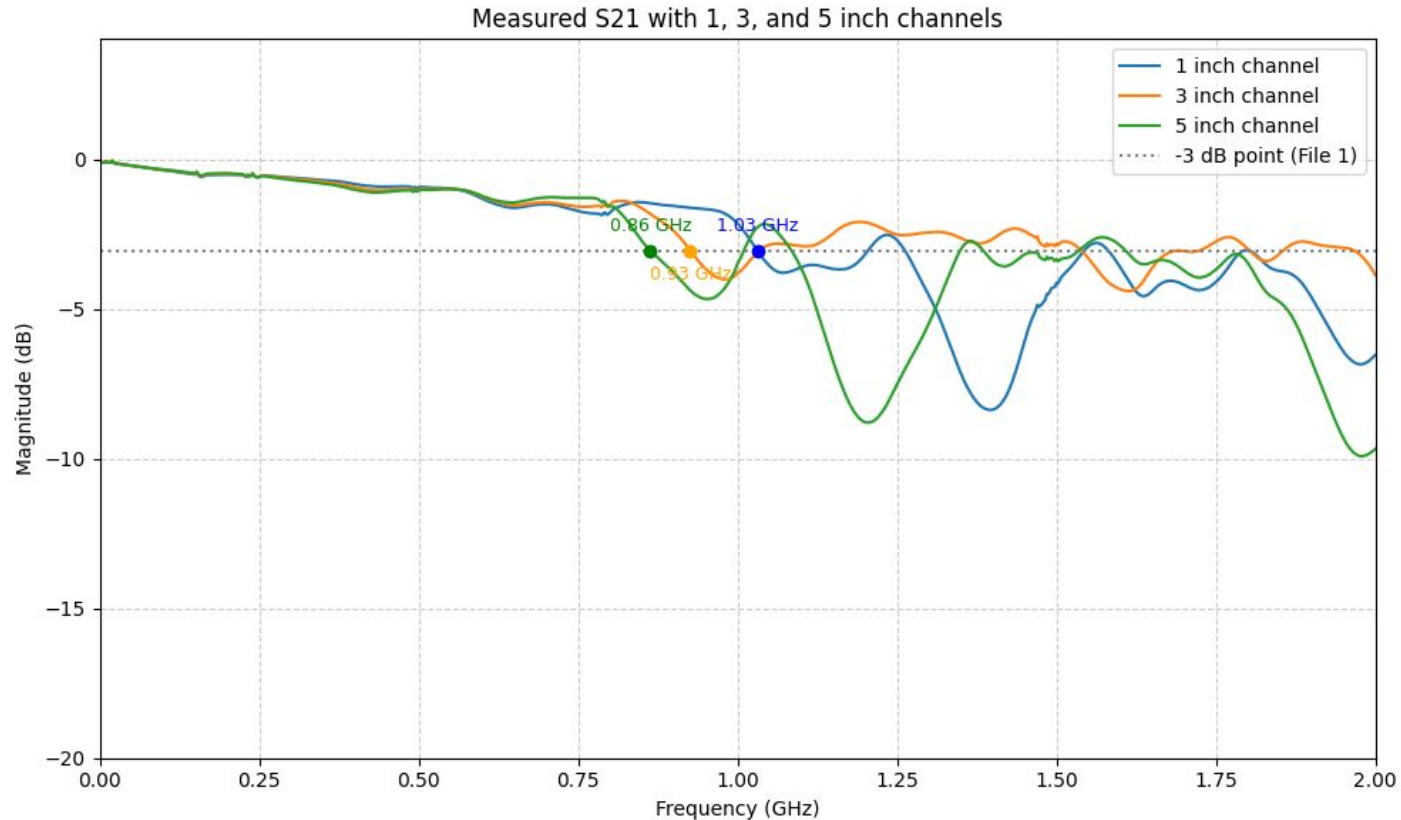
Parasitic Cap and Vias with and without T-coils (Measured)



1, 3, and 5 inch Channels (Simulated)



1, 3, and 5 inch Channels (Measured)



Conclusion

- Calculated, modeled, fabricated, and characterized T-coils to address parasitic capacitance and expand channel bandwidth
- T-coils successfully improved bandwidth and helped compensate for capacitance
- Measured data demonstrated additional losses compared to simulation results
- Gained experience working with various design tools (Keysight ADS, Ansys HFSS, HyperLynx) and lab characterization with VNA
- Future work:
 - Based on HFSS simulations, our design maybe has a higher L than the ideal ones based on the formulas. A higher C_L may be better tuned to match the impedance
 - We could also explore other inductor topologies to more finely tune L

References and Acknowledgments

References

- [1] B. Razavi, "The Bridged T-Coil," *IEEE SOLID-STATE CIRCUITS MAGAZINE*, vol. 7, no. 4, pp. 9–13, Fall 2015.
- [2] M. Xiong, "ESD Fundamentals, Part 4: ESD Capacitance." ti.com.
https://www.ti.com/lit/ta/sszt796/sszt796.pdf?ts=1777869974857&ref_url=https%253A%252F%252Fwww.google.com%252F (accessed May 3, 2026).
- [3] J. Cañada, J.M. López-Villegas, N. Vidal, S.F. Nagle, and L.F. Velásquez-Garcísquez-García, "Geometry optimization of broadband, planar, spiral inductors via continuous trace width modulation," *Sci Rep*, **15**, 38728, Nov, 2025.
- [4] M. T. Reiha and J. R. Long, "Symmetric Monolithic T-coils for Broadband IC Design," *2007 IEEE/MTT-S International Microwave Symposium*, Honolulu, HI, USA, 2007, pp. 1909-1912, doi: 10.1109/MWSYM.2007.380146.

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